

Draft - expert review pending

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Topic 1.3 - Kinds of computers

A real classroom situation

Mr. Waqar runs the computer lab at a higher secondary school in Sukkur. One morning a pupil points around the lab and asks: "Sir, the office has a big box under the desk, the head sir carries a folding screen, you keep a small black box in the cupboard with lights on it, and my cousin says her bank has computers the size of cupboards. Which one is the real computer?" Mr. Waqar smiles, because the question has a real answer: they all are, and the differences between them are exactly how computer scientists sort them into kinds. Being able to sort them, he knows, is the day's lesson.

Explanation

Classification of computers by functionality: analog, digital, hybrid

One way to sort computers is by the kind of signals they work with.

A digital computer works with discrete values, in practice the 0s and 1s of the binary system. Every number, letter, photo and sound is stored as patterns of these two states. The laptop, the phone and the lab desktop are all digital computers, and nearly every computer you will meet is digital.

An analog computer works with continuous physical quantities, such as voltage or the rotation of a shaft, that change smoothly the way temperature or speed do. A classic example is the slide rule, and analog computers were once used to aim naval guns and simulate flight in the mid-1900s. An analog machine does

not count; it measures. Where a digital computer answers "exactly 7", an analog computer answers "about there, on this dial".

A hybrid computer combines both: it measures an analog quantity, converts it into digital form, and processes it digitally. Hospital patient monitors do this: they continuously measure a heartbeat as an analog signal, then convert it to numbers for display and alarms. Modern petrol pumps do the same with fuel flow.

Three ways computers handle information

Compare the kind of signal, the method, and a familiar example.

Type	Signal it works with	How it answers	Example
Analog	Continuous physical quantity	By measuring	Slide rule
Digital	Discrete 0s and 1s	By counting exactly	Laptop and smartphone
Hybrid	Measures analog, then converts to digital	By measuring, then counting	Hospital patient monitor

A hybrid computer links a measured signal with exact digital processing.

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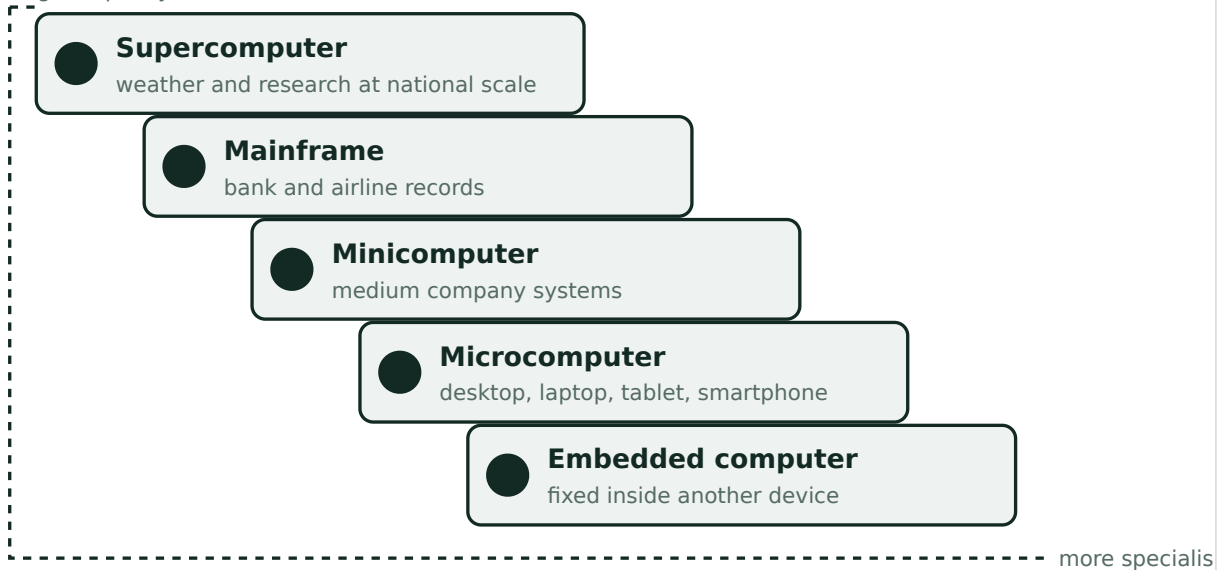
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Classification of computers by size and power

The second way to sort computers is by size and processing power. Introductory texts such as Norton's *Introduction to Computers* and Shelly and Vermaat's *Discovering Computers* (cited at bibliographic level as the course guide lists them) describe the same ladder, from the most powerful to the smallest.

Computer classes, size and power

Larger shared systems at the top, smaller specialised systems at the bottom
larger capacity



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A **supercomputer** is the fastest class of machine, used for weather forecasting, climate research and scientific simulation. Pakistan's research and meteorology communities use supercomputing time for weather and earthquake modelling.

A **mainframe** is a large machine built to process enormous, steady streams of records for banks, airlines, telecoms and tax departments, serving thousands of users at once. When a bank in Karachi posts a million transactions overnight, a mainframe or mainframe-class server is at work.

A **minicomputer** (also called a mid-range server) serves a medium organisation or one department: a university's examination records, a hospital's stores, a factory's line. The "classic" minicomputer is mostly history, but the mid-range server role continues.

A **microcomputer** is a computer whose entire processor fits on a single chip, the microprocessor. The desktop, the laptop, the tablet and the smartphone are all microcomputers. This is the class a teacher works with daily, and the phone is now the most common microcomputer in Pakistan.

An embedded computer is a small computer fixed inside another device to run that one device: the controller in a washing machine, a car's engine

management unit, an ATM's inner computer, a biometric attendance machine. You do not use it as a computer; it works for you inside something else.

A caution about the ladder: it is about size and power of the machine, not importance. The embedded controller in a hospital's infusion pump matters far more to a patient than a supercomputer forecasting next month's weather. And the classes keep sliding: today's phone outpowers yesterday's mainframe. Classify by what the machine is built to do, not by how old it looks.

Activity: Device detective

Work in pairs. About 20 minutes.

1. Your teacher will describe six devices, one at a time: an ATM, a smartphone, a national weather-forecasting system, a school office desktop, a car's engine controller, and a hospital patient monitor.
2. For each device, agree with your partner on two labels: its functionality class (analog, digital, or hybrid) and its size class (supercomputer, mainframe, minicomputer, microcomputer, or embedded).
3. Write your answer as "device: functionality class + size class, because ...", where the "because" names one feature of the device that proves the label.
4. Compare with another pair. Any device you labelled differently, discuss until you can agree on which feature settles it.

Check your understanding

1. State the difference between an analog computer and a digital computer in one sentence, with one example of each. (*Remember*)
2. Why is a hospital patient monitor classified as a hybrid computer? Answer in two sentences. (*Understand*)
3. A device is fixed inside a car and controls fuel injection. It never shows a screen and the driver never operates it directly. Classify it on both axes and justify your answer. (*Apply*)
4. Explain why a smartphone is a microcomputer even though it fits in a pocket. (*Understand*)

Summary

Computers are sorted on two axes. By functionality: analog computers measure continuous quantities, digital computers count discrete 0s and 1s, and hybrid computers measure then count. By size and power: supercomputers, mainframes, minicomputers, microcomputers and embedded computers, from national-scale simulation down to the controller hidden inside a washing machine. Almost every computer you will use as a teacher is a digital microcomputer, but knowing the whole ladder lets you place any device you meet.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define analog, digital and hybrid computers with one example of each.
- ☐ I can name the five size classes of computers in order of power.
- ☐ I can classify a described device on both axes and justify the label with a feature of the device.

Try this at your practicum school

Walk through your practicum school and list every computer you can find, including hidden ones (the biometric attendance machine, the photocopier, the UPS controller). For each, note its size class and what it is used for. Aim for at least five devices.

Summative task

A school receives a donation of five devices: a desktop computer, a tablet, a biometric attendance machine, an old bank-server rack, and a weather receiver built to process continuous analog signals into digital forecasts. Prepare a classification table with one row per device and three columns: functionality class, size class, and one sentence of justification naming the feature that settles each label.

Marking guide (10 marks):

- Functionality class correct for all five devices: 3 marks.
- Size class correct for all five devices: 3 marks.
- Justifications name a defining feature of the class rather than restating the label (Analyse): 3 marks.
- Table is clearly organised: 1 mark.

Further reading

- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 2. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Norton, P. *Introduction to Computers*. McGraw-Hill Education. (Course guide reading; print monograph, cited at bibliographic level.)
- Khan Academy. *Computers and the Internet*.
<https://www.khanacademy.org/computing/computers-and-the-internet>

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